Gestão de Inovação e Design Thinking/Innovation Management and Design Thinking 2026/27

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Week 4

- Inspiration through empathy and qualitative research

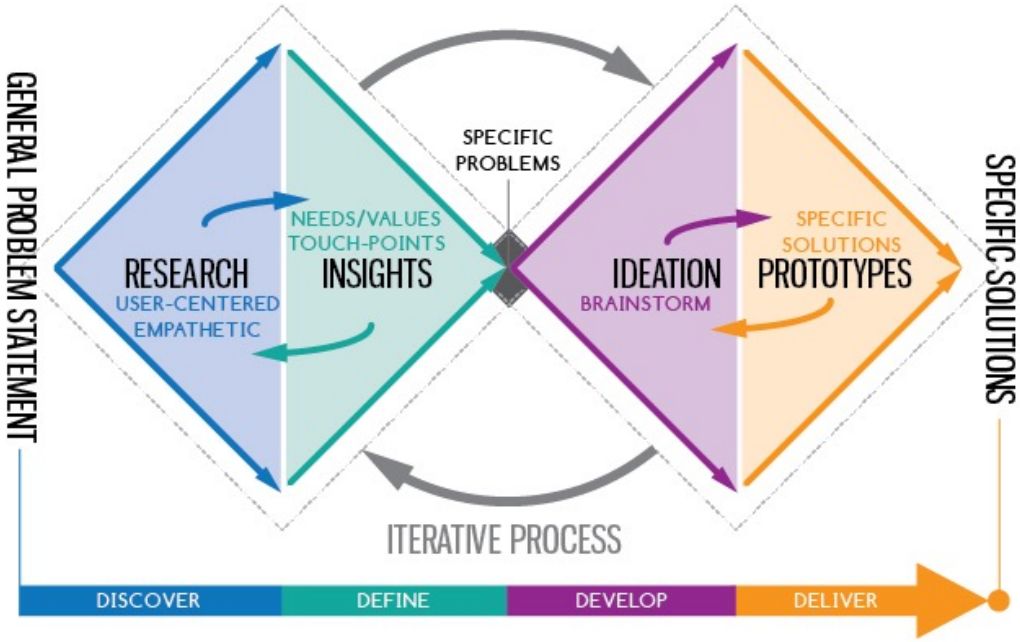
Design Thinking Projects – The problem

How can we encourage youth (aged 14 - 18) to participate and intervene in urban mobility solutions?

RESEARCH DESIGN & OBSERVATION TOOLS



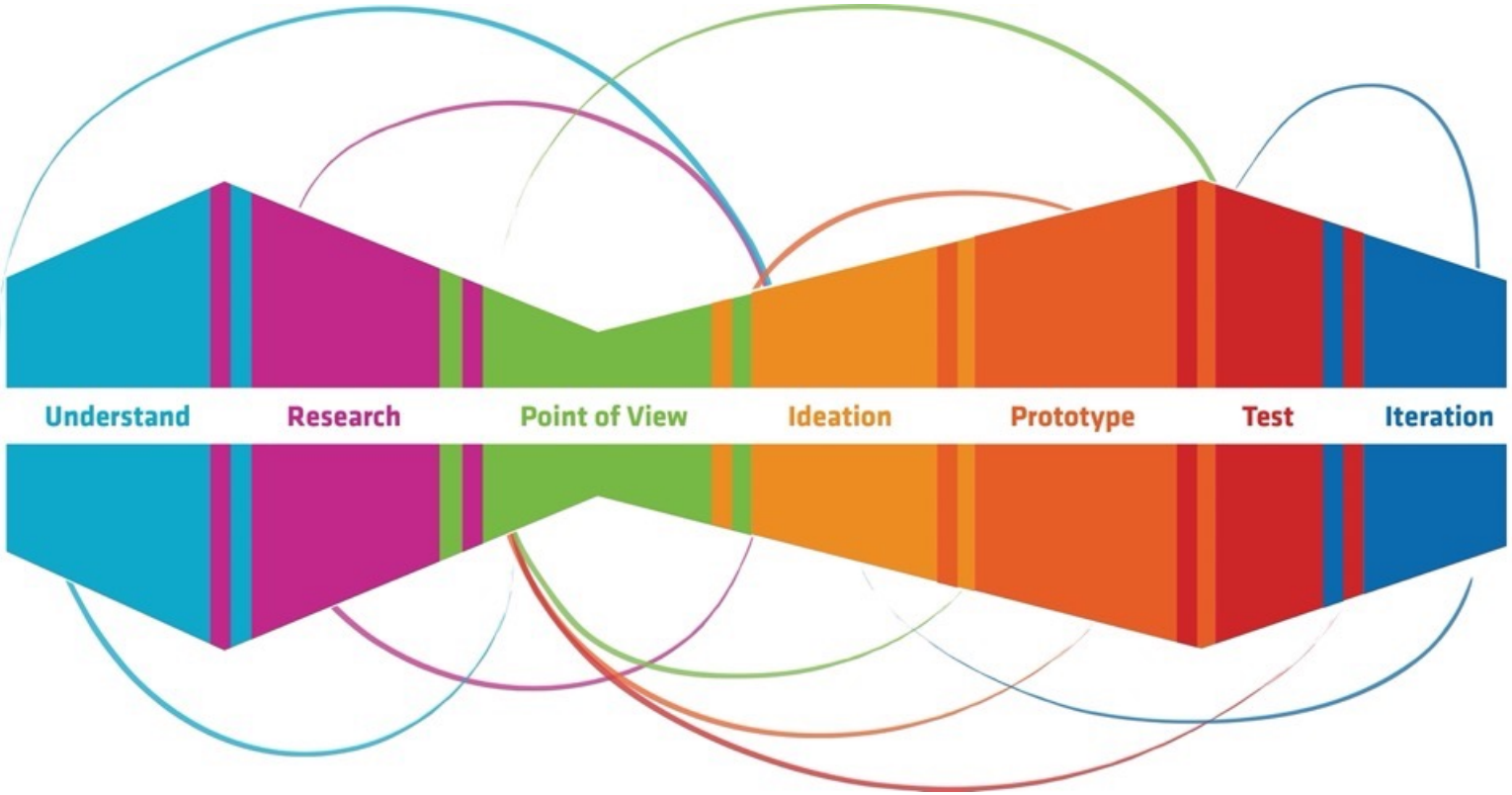
Double Diamond DESIGN PROCESS



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Service Design Vancouver
building innovation + value for businesses and people

THE PROCESS


Understand
Research
Point of View
Ideation
Prototype
Test
Iteration

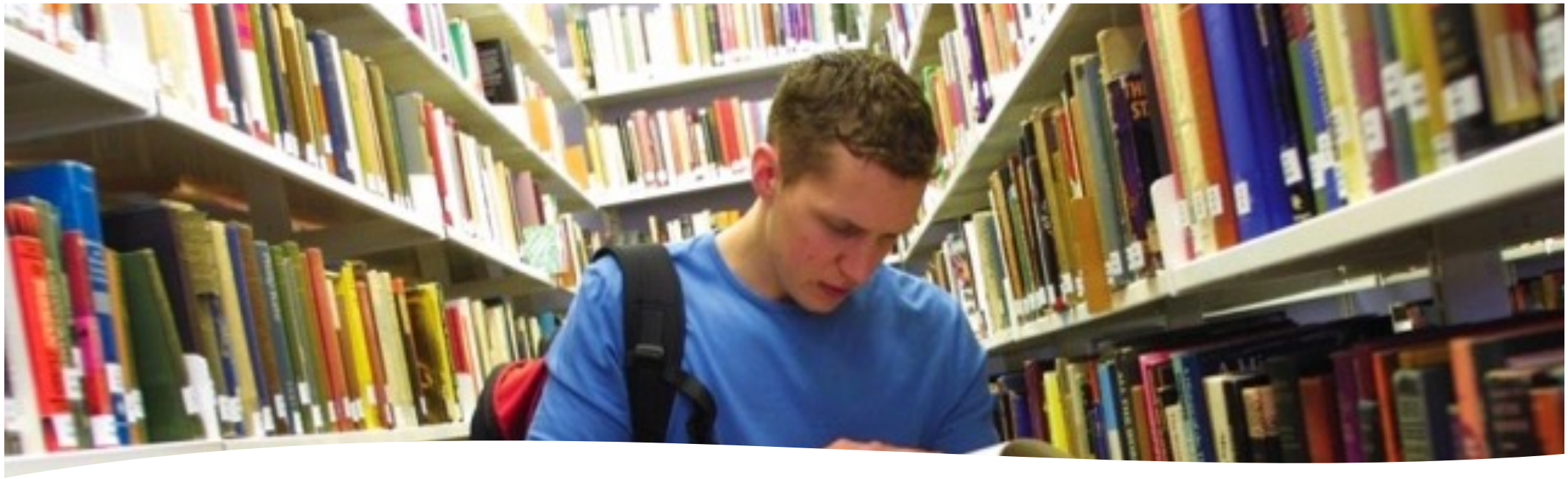
1- Inspiration

- Understanding the challenge
- The aim is to get to the bottom of the initial problem through various sources and methods, including, for example, interviews and documentary research, and observation and empathy tools to gain insights.

ASSUME A BEGINNER MINDSET

- Don’t judge
- Question everything
- WHY? WHY? WHY?
- Be truly curious
- Find patterns
- LISTEN





UNDERSTAND
You need to become an ‘expert’

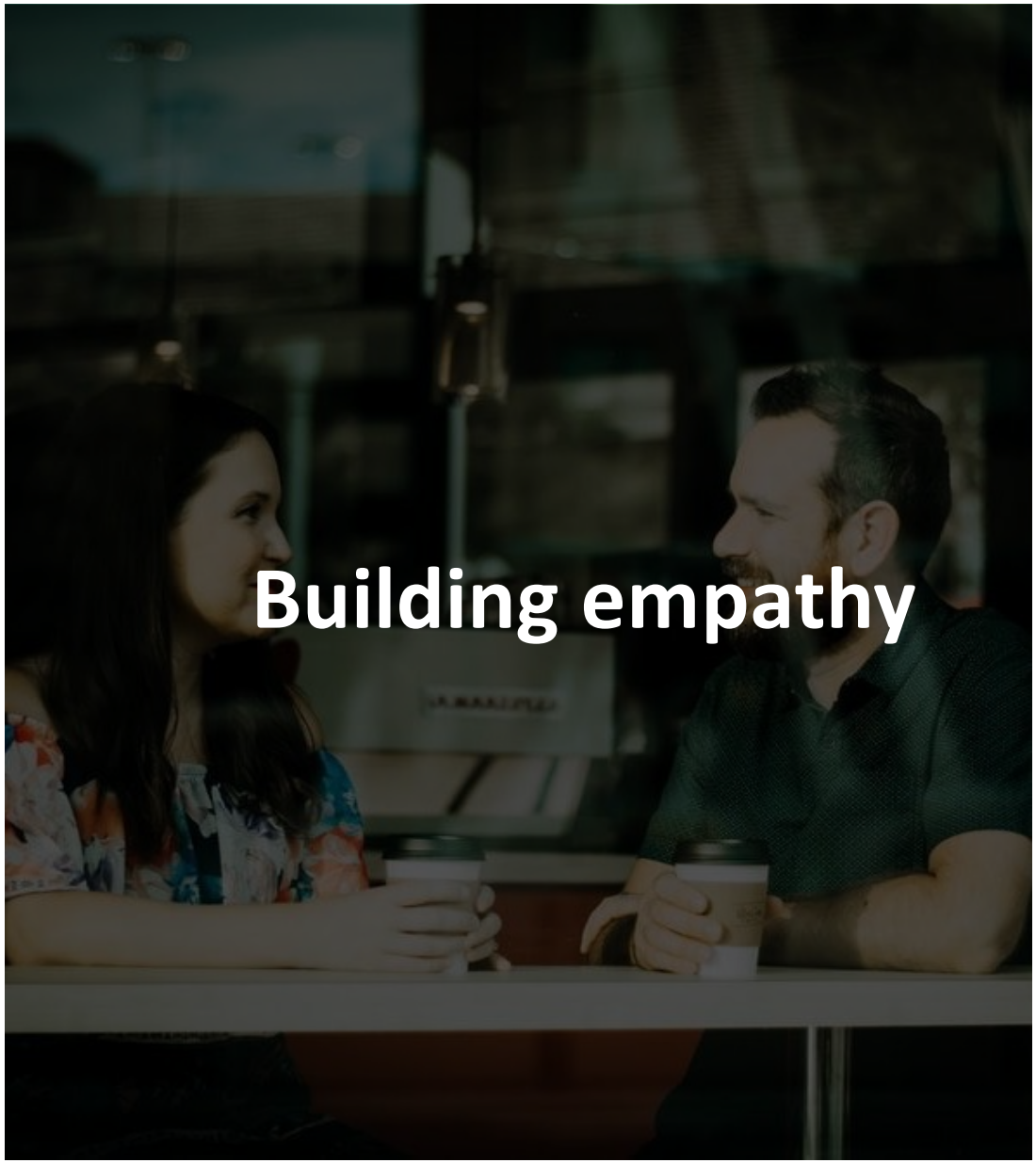
- **Get a shared understanding of the problem**:
  - Share stories about your own experiences
  - Collect assumptions
  - Discuss existing solutions
  - Research about the topic
- **Get motivated!**



- YOU CAN NOT SOLVE A PROBLEM
- IF YOU NOT EXPERIENCE TO BE PART IT

RESEARCH

- STEP INTO THE USER'S SHOES
- UNDERSTAND THEIR WORLD.
- GAIN INSPIRATION FOR INNOVATION
- DISCOVER NEEDS AND PROBLEMS
- GET FACTS



UNDERSTAND PEOLPES' FEELINGS



OBSERVE DIFERENT PERSPECTIVES



- LOOK FOR THINGS THAT DON’T WORK



LOOK FOR SURPRISES



LOOK FOR ANALOGIES

- Analogies can be a powerful tool to find insights that are not obvious to direct approach.
- Comparing with other companies that offer similar products, services or experiences.
- Try to find out where are the better and the worst results.







- LOOK FOR IMPROVISED SOLUTIONS





- LOOK FOR EXTREME USERS


DEEP OBSERVATION



WHAT? HOW? WHY?

- **Describe in detail what the person that you are observing is doing.**
- **This tool helps you to look what´s around you in a different way, you start to see with attention.**

PEOPLE ARE ALL DIFFERENT



- The way they feel
- The way they see
- The way they hear is different





Our colleague Kristian admitted himself as a patient, he saw the dull ceilings all day and heard medics speaking in acronyms. This became an experience he could design against. *Do you live like the customer?*

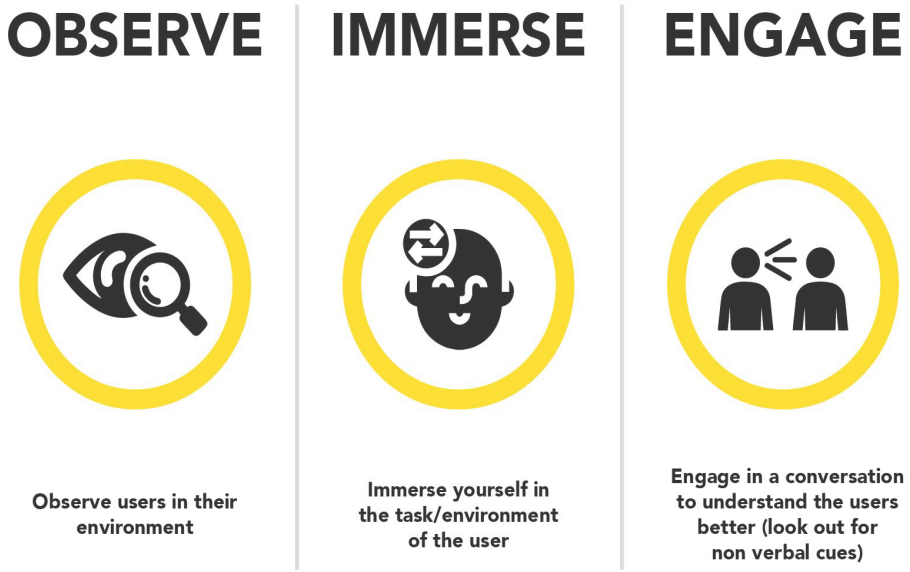


What is the Empathize Mode?

- Empathy is the ability to step into other people's shoes, to understand their lives and start to solve problems from their perspectives.
- To understand users, we need to talk to them.
- Talking to users is difficult and can make things go from bad to worse if it's not done properly
- When we talk to users, we need to ask them GOOD questions
- Start the empathy work by reaching out to people who maybe have a different perspective from you because that's going to challenge your own understanding of "How might we" assumptions.

Empathy is the foundation of human-centred design process necessary to:

- **Observe**: View users and their behaviour in the context of their lives
- **Engage**: interact with and interview users through both scheduled and short “intercept” encounters
- **Immerse**: Experience what your users experience



Empathy Map

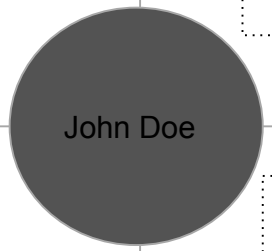
NB: Goals/motivations/desires

- Contains what the user said during our conversation
- Should be reported verbatim

SAYS

- Captures what the user is thinking throughout the experience
- Contains details of what matters to the user

THINKS



- Captures the actions of a user
- Captures how the user carries out the action

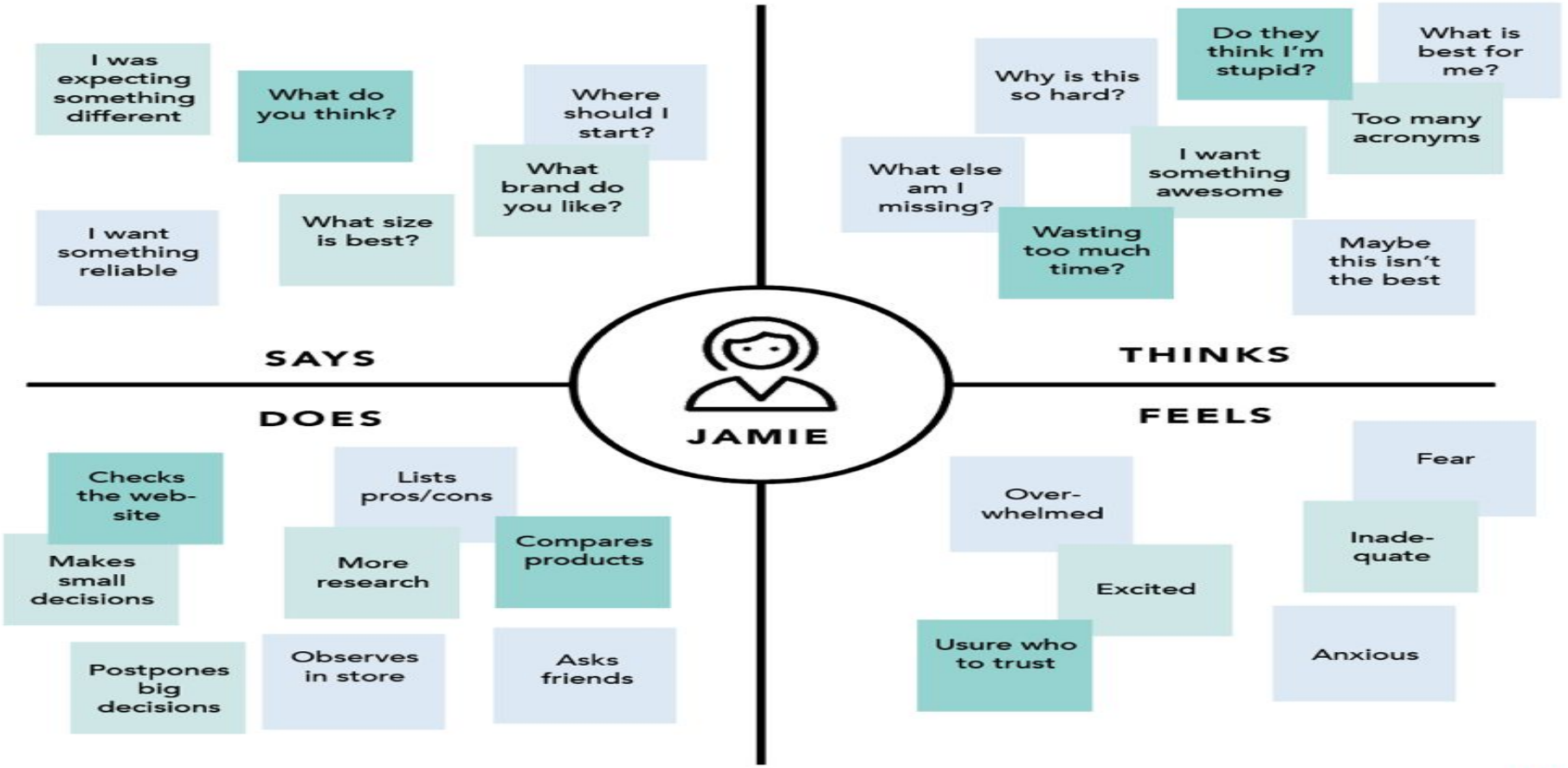
DOES

- Captures the emotional state of the user
- Captures details of how the user feels about the experience; what worries the user; what excites the user

FEELS

NB:Body language/tone of voice

EMPATHY MAP *Example (Buying a TV)*



[Design Tool]

Storytelling

Put the human beings and their experience front and center. Take the listener along on their journey.



UNDERSTAND THE PROBLEM

- What's the REAL problem?
- What are the persons' needs?
- Where can you find disappointment? Unhappiness?
- Restrictions?
- What is important to understand?
- What do we want to know about?

- Hielata, K. (2019, November). Complementing the scoping process of configuration projects by design thinking. In *Transdisciplinary Engineering for Complex Socio-technical Systems: Proceedings of the 26th ISTE International Conference on Transdisciplinary Engineering, July 30–August 1, 2019* (Vol. 10, p. 33). IOS Press.

Design thinking phases	Scoping and KM in configuration projects	Design research	Suggested methods and tools
Exploring the problem space	Determining the scope of the project	Seek to understand culture, experiences, emotions, thoughts, and behavior: What are the user experiences? What will make them happy in achieving goals?	Applying qualitative research methods such as ethnography, participant observation, interviews, journey mapping, job-to-be-done analysis, mind mapping to facilitate drawing insights from ethnographic data and to create a "common mind" across team members, using collaborative ideation such as brainstorming and concept development techniques
	Knowledge acquisition	Primarily through semi-structured conversations between the researcher and the interviewed. Definition of integrations, architecture and user interface.	
Exploring the solution space	Development and knowledge validation	Represents samples qualitatively and seeks profiles of extreme users, as unusual observations may lead to novel ideas	Identifying assumptions: assumptions around value creation, execution, scalability, and defensibility that underlie the attractiveness of a new idea Prototyping: techniques that facilitate making abstract ideas tangible (storyboarding, user scenarios, metaphor, experience journeys, business concept illustrations, and so on)
	Documentation and knowledge maintenance	Investigation on behavior, objects and words that people use to express the way they interact with the system	

Tabela 1. Exemplo de técnicas a utilizar em cada fase, de acordo com os objetivos

Fase	**Objetivo**	**Exemplo de técnicas a utilizar**
Inspirar	Criar empatia com os utilizadores, compreender as necessidades dos utilizadores, formular problemas de conceção	Pesquisa exploratória (*benchmarking*, análise de tendências, universos paralelos, pesquisa bibliográfica), pesquisa etnográfica (entrevistas, mapeamento do percurso do utilizador), visualização (captura de ideias e conceitos num quadro branco, mapeamento mental) e síntese (encontrar padrões na pesquisa, identificar *personas*, formular ideias).
Idealizar	Aplicar técnicas de pensamento criativo e divergente para gerar e aperfeiçoar ideias	*Brainstorming*, técnicas de pensamento divergente, visualização (captura de ideias e conceitos num quadro branco), priorização de ideias (pontuação e seleção de ideias para implementação com base na sua conveniência, exequibilidade e viabilidade).
Implementar	Prototipar e testar pressupostos	Prototipagem (*storyboards*, maquetas digitais ou físicas), testes com utilizadores (síncronos e assíncronos).

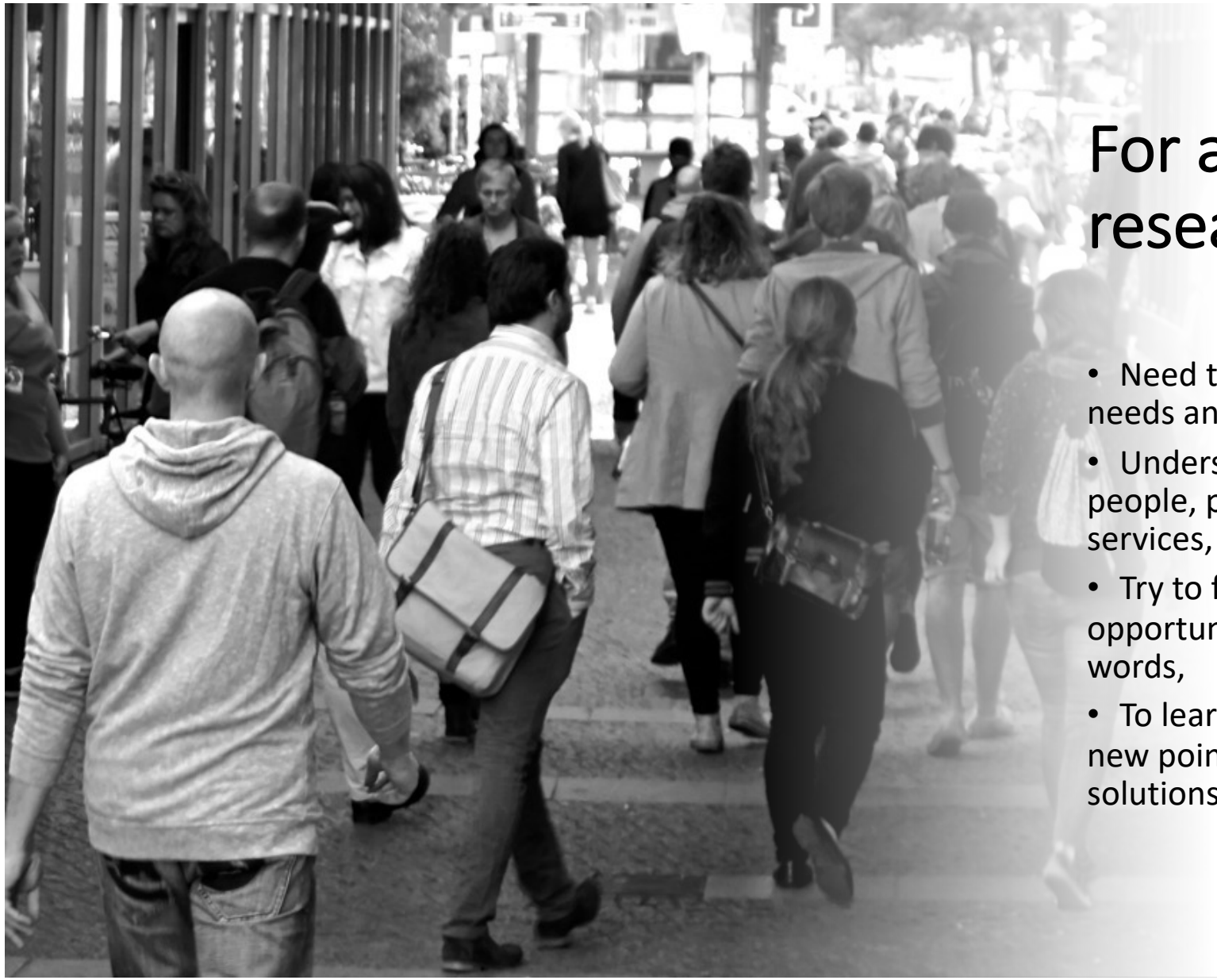
Fonte: adaptado de Victorino (2023)

METHODS

- Extreme Users
- Paralel Universes
- Emphatic Experiences
- In Depth Interview (users, non users, experts)
- Day in a life
- Customer Journey
- Netnography

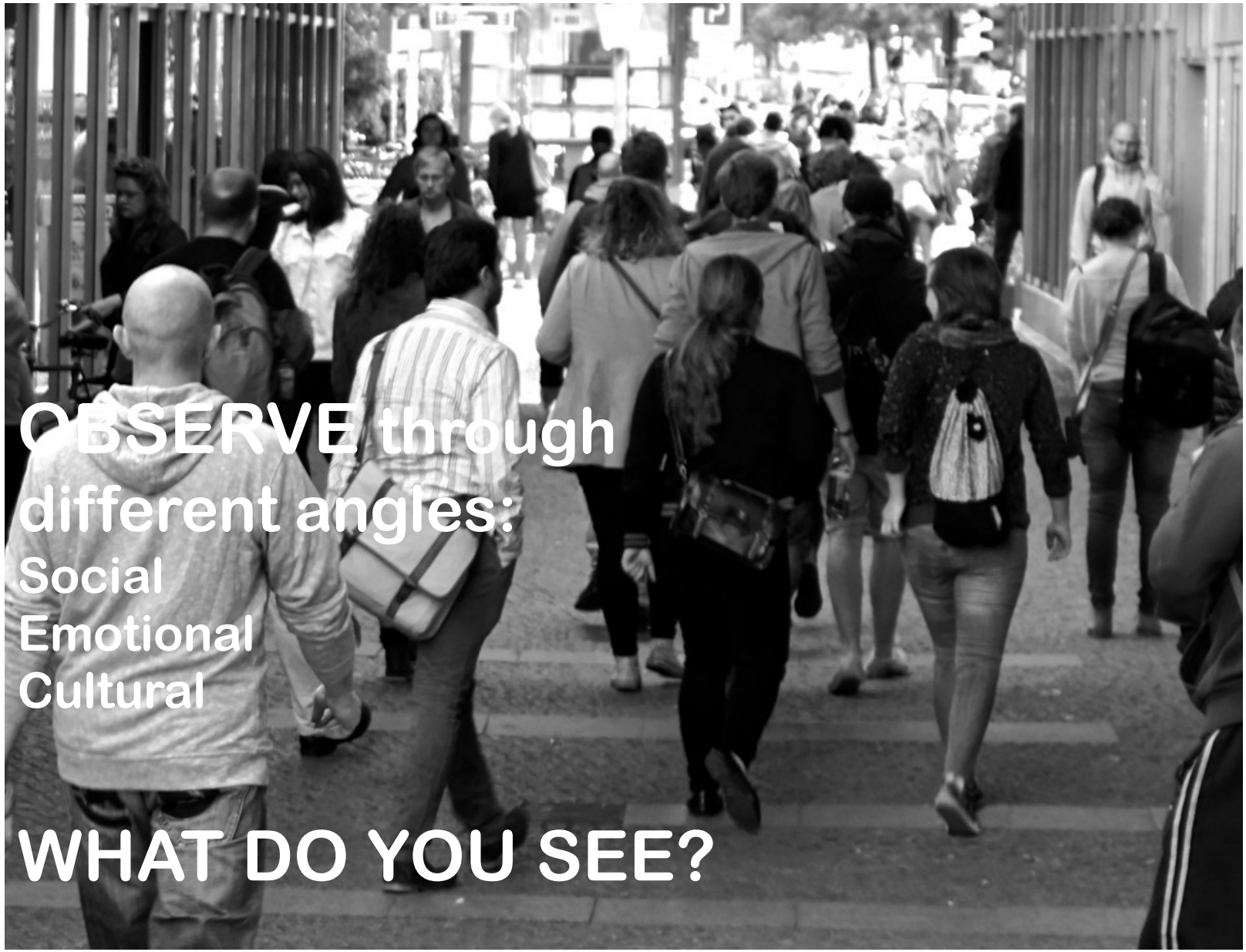
METHODS

- Parallel worlds – other contexts from which we can make parallels, and adapt to our context
- Extreme users - allows you to find inconsistencies, their behavior is more prominent and therefore easier to identify. The hard part is sometimes finding the extreme users (the ideal is to observe the two ends of the spectrum in pairs).
- A day in the life - ethnographic technique - I ask someone to describe their day to me, with photos and comments, recording their frustrations or the things that don't work.
- Fly on the wall - passive observation
- Empathic experience - we impersonate the client
- Ways of obtaining information: Verbal (interviews); non-verbal (observation)



For a qualitative research:

- Need to understand people needs and desires,
- Understand relation between people, places, objects and services,
- Try to find out barriers and opportunities based on users words,
- To learn with all the answers a new point of view for ideas and solutions

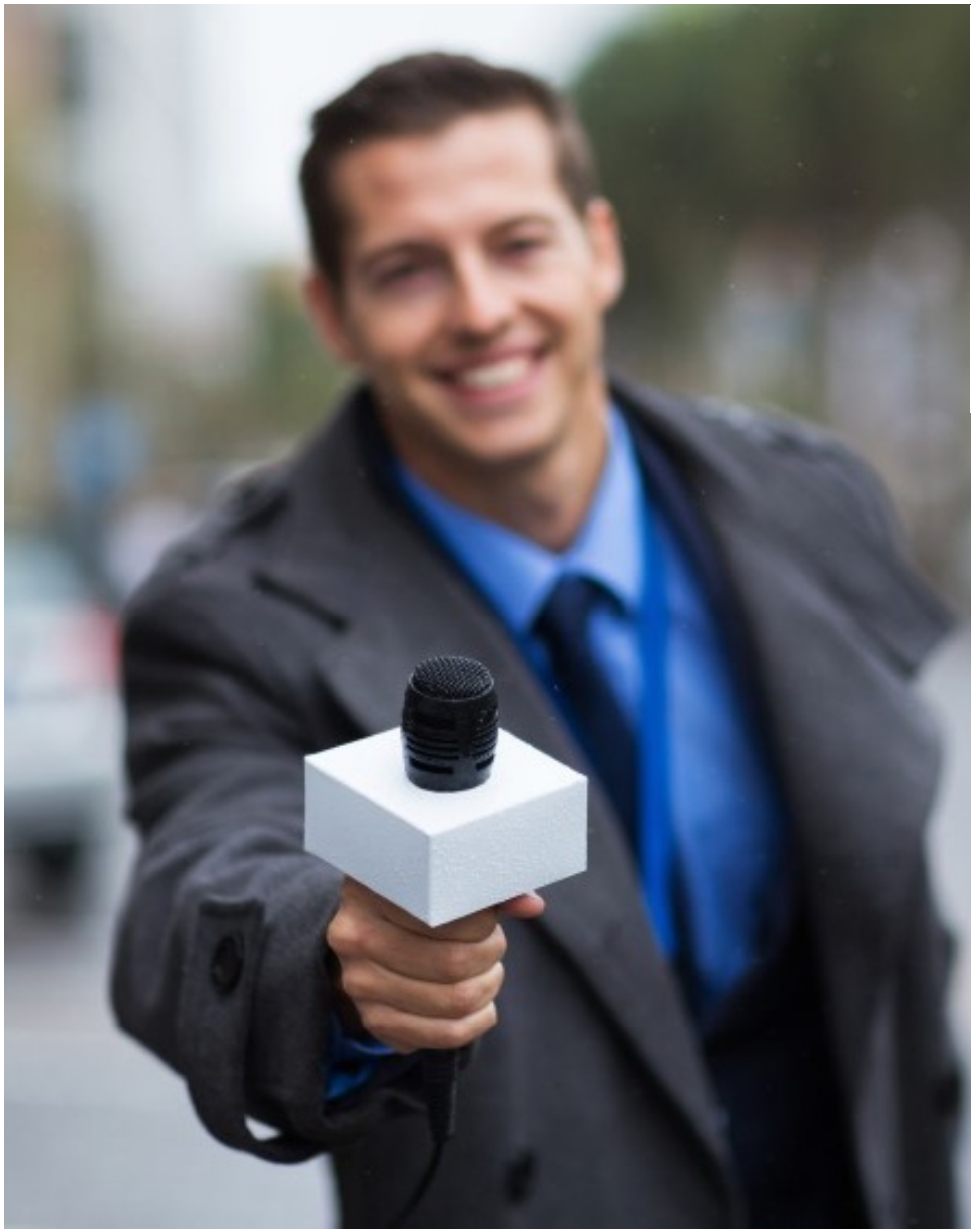


LOOK FOR SURPISES

Be alert and curious

Do not judge

Look as a traveller



- INTERVIEWS
- Build empathy and collect
- life experiences

- LISTEN to new points of view,
- ideas and suggestion

- Listen 80%
- Talk 20%

- Try to find EXREME USERS

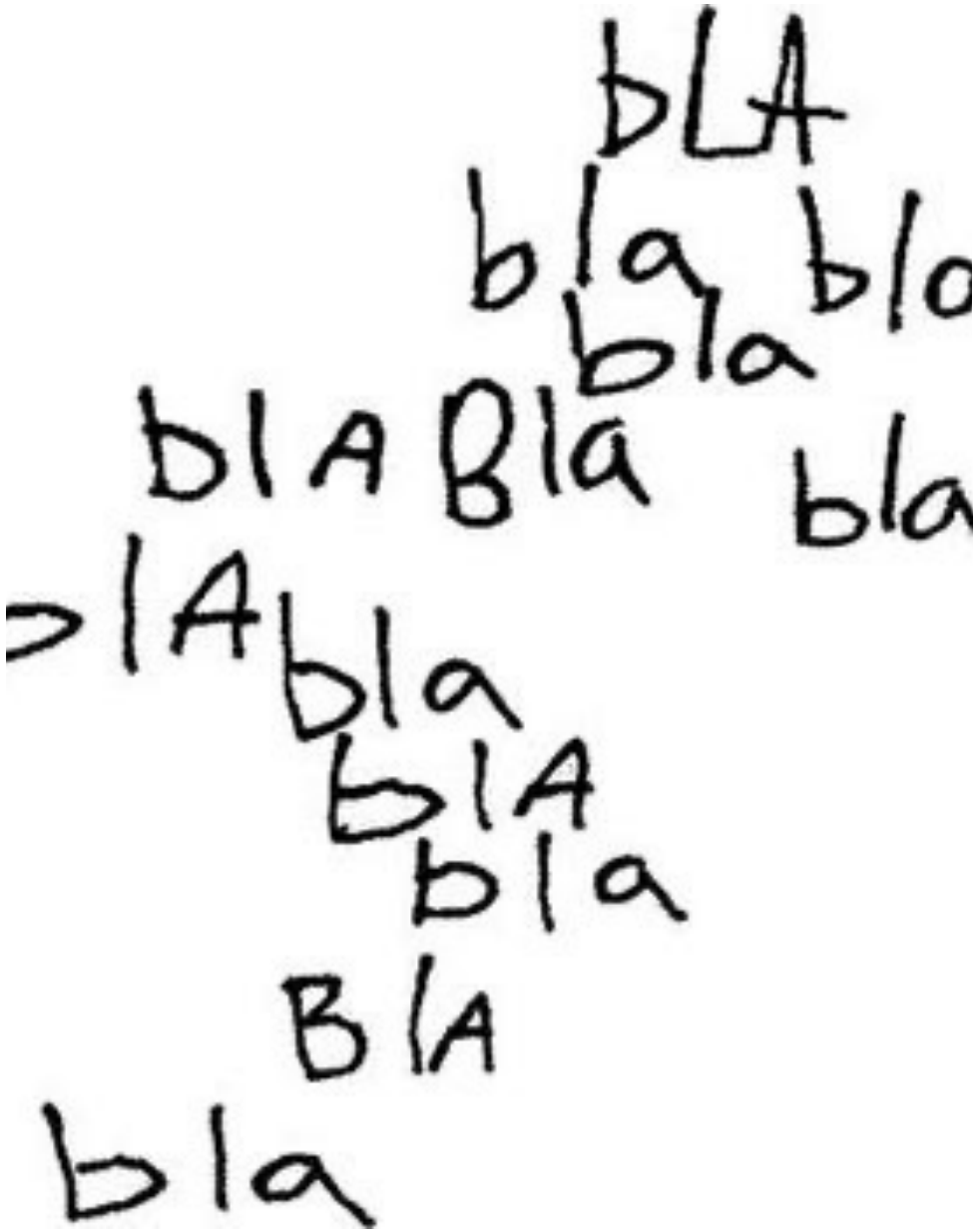
We all have our own listening framework

- Culture
- Language
- Values
- Beliefs
- Attitudes
- Expectations
- Intentions
- …



Cognitive Bias





- WHEN WE LISTEN,
- WE RETAIN …%
- OF THE INFORMATION



- We retain just 25% of the information

Interview for EMPATHY

1

To get information from people that we don’t know at all, building empathy and a deep interaction is not a easy task.

2

Time with users is precious, you need to make the most of it!

3

You may not get to every question you prepared, but you should prepare a good engagement plan.

INTERVIEW Preparation

- Prepare a list of questions without loosing the focus: you are looking for emotions, pains, motivations, and most of all, the way people react to the service or product in cause.




It helps to make a CHECK LIST
with topics and questions
you want to address

Interview Preparation

Don’t forget to introduce yourself and the propose of the interview



Interview Preparation

- **Start with an open and generic questions, humor is important. Observe the reactions, the expressions the body language.**



Interview Preparation

- Try to stimulate the imagination with different scenarios and environments
- ASK IF?



Interview Preparation

- Try to dig more into the information
- that is most relevant
- ASK WHY?



- What do we want to know?
- What do we…
- SEE? Observe what people is doing
- HEAR? Hear what people say and do
- FEEL? Do what people do



- **DON’T FORGET**
- **Record the interview**
- **Take notes**
- **Take pictures**





- **GO FOR THE INTERVIEWS MORE INFORMATION YOU GET BETTER SOLUTIONS YOU CAN DESIGN**

PERSONAS

- A persona is a fictional, yet realistic, description of typical or target user created based upon user research in order to represent the different user types that might use your service, product,site, or brand in a similar way.
- Creating personas helps to understand your users' needs, experiences, behaviours and goals.
- Personas make the design task at hand less complex as they guide the ideation process, and ensures that your target user is kept in focus through all the design and development process.

Types of Personas

- Goal Oriented persona - examines the process and workflow that your user would prefer to utilise in order to achieve their objectives in interacting with your product or service (function)
- Role based Persona - focuses on the user's role/function and behaviour (context of the product)
- Engaging persona - examines the emotions of the user, their psychology, backgrounds and make them relevant to the task in hand.
- Fictional persona - based on assumptions of the team upon past interactions with the user base, and products to deliver a picture of what, perhaps, typical users look like. Should be used as aninitial sketch of user needs and NOT as a trusted guide.



Rebecca

Casual audiophile

Age	26
Occupation	Frontend developer
Education	Bachelor degree
Marital status	Single
Location	Mountain View
Online locations	Work and mobile
Computer(s)	iPhone and MacBook Pro
Internet usage	8-9 hours

TECHNOPHOBE — TECH WIZ

CDs — MUSIC STREAMING

CASUAL LISTENER — HARDCORE GEEK

Music is essential to Rebecca's life. She is listening to tunes almost every second of her life, particularly while working.

Obstacles Rebecca faces:

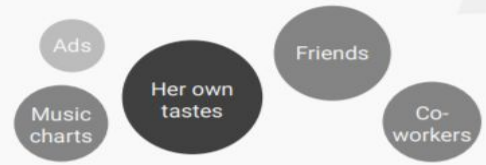
- Too busy to explore new music artists she might like
- Streaming music consumes a lot of data

How will Rebecca interact with Spotify?

Questions Rebecca will ask:

- How do I keep updated on new releases by artists I follow?
- How do I learn of new artists I haven't heard of?
- Can I listen to music in a data-efficient manner?
- How can I listen on both my MacBook and my iPhone?

Who influences Rebecca?



Rebecca's situation

Goals, motivations:

- Listen to great music to keep her productive at work
- Relax and unwind at the end of the day
- Superior music quality for full enjoyment of tracks
- Expand the circle of music artists she listens to

Key words

music, jazz, r&b, pop, artists, new releases, top charts, background music

Rebecca's story

Music is a big part of my life; I like to think that I always have a "background music" running in each scene of my life. I love working while listening to music; somehow, it gives me a lot of focus on my task.

I regularly talk to my co-workers about music and singers — that's what we like to talk about over lunch. We're constantly looking for new artists to inspire us and to expand our music library, but lately it seems a little tough to do that. Everything seems to have a "filter bubble" effect, and we keep listening to the same genres and artists.

I really enjoy finding new artists that match my subjective taste, and most of the times I get those from my close friends. I wish there were a way to find more music and artists without having to rely on the serendipity of life!

Example of a persona that shows the six main elements you should include. Name, age, gender, tag line, experience and skills are placed on the left-hand side. The middle column focuses on the context to indicate how they would interact with a product or service. Finally, on the right-hand side some goals and concerns are shared, as well as a short scenario to indicate the persona's attitude.

Jill Anderson



"I'm looking for a site that will simplify the planning of my business trips."

AGE	29
OCCUPATION	Regional Director
STATUS	Single
LOCATION	Portsmouth, NH
TIER	Frequent Traveler
ARCHETYPE	The Planner

Organized	Practical
Protective	Hardworking

Bio

Jill is a Regional Director who travels 4-8 times each month for work. She has a specific region in which she travels, and she often visits the same cities and stays in the same hotel. She is frustrated by the fact that no matter how frequently she takes similar trips, she spends hours of her day booking travel. She expects her travel solutions to be as organized as she is.

Personality

Extrovert	Introvert
Sensing	Intuition
Thinking	Feeling
Judging	Perceiving

Brands

KAYAK, Basecamp, Outlook, enterprise, IHG InterContinental Hotels Group

Goals

- To spend less time booking travel
- To maximize her loyalty points and rewards
- To narrow her options when it comes to shop

Frustrations

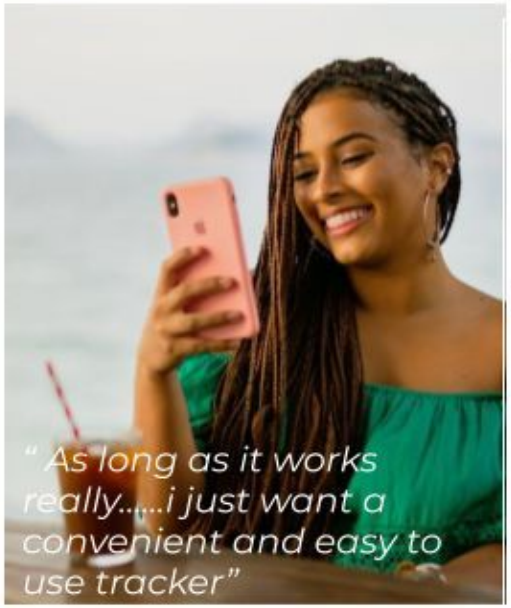
- Too much time spent booking - she's busy!
- Too many websites visited per trip
- Not terribly tech saavy - doesn;t like the process

Motivations

- Price
- Comfort
- Convenience
- Speed
- Preferences
- Loyalties/Rewards

Technology

- IT & Internet
- Software
- Mobile Apps
- Social Networks



Amaka Eno

25

Customer Service Executive

#Ambitious #outgoing #planner #fitfam #adulting

Tech Savvy: ★★★★☆

GOALS

- To live a healthy lifestyle
- Wants to a Financial Adviser
- Earn enough to not worry about bills and afford vacations.

NEEDS

- Accurate Ovulation and menstruation tracking
- Log lifestyle habits
- Health insights and tips

PAIN POINTS

- Forgets to log symptoms, moods etc.
- Feels insecure and self conscious when she is on her period

BEHAVIOUR

Amaka is a feisty go getter. She works long hours and always looks forward to the weekend when can sleep in, netflix and chill. She enjoys evening hangout with friends and is dating casually. She doesn't love her current job and hopes to transition to financial sector, to make this happen faster, she is taking online courses to develop her skills. She recently got an apartment with a friend and hates BILLS. She wants to incorporate healthy habits in her life but seldom has time to.

TECH HABITS

- Apple addict
- Has wifi at home
- Has a fitbit

What do we already know about our design challenge?

What we don't know about our design challenge?
(or we would like to know more of…)

- What do people need or want?
- What technologies can help in this challenge?
- What solutions or ideas are being tried in other areas?
- What early hypotheses have your formed about how to solve your design challenge?